

Week 7 (7.3-7.9)

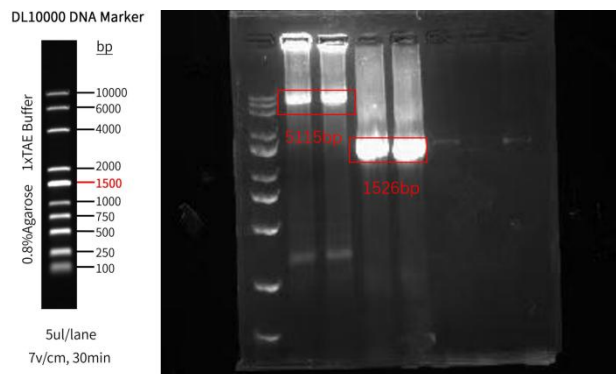
Module 1:

(A)Construct BW-pYB1a-eGFP-progo-nov1f

1.PCR + cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and vector pYB1a. Subsequently, the relevant DNA fragments were assembled using Gibson Assembly methodology, followed by transformation into competent cells. Upon completion of the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed.

PCR system (50μL)		PCR-pYB1a-progo		} × 30
2×Mix	25μL	98°C	5min	
PROGO-F	2μL	98°C	30s	
PROGO-R	2μL	64°C	30s	
Template	20ng	72°C	2min33s	
DDW	20μL	72°C	5min	
		25°C	∞	
PCR system (50μL)		PCR-nov1f		} × 30
2×Mix	25μL	98°C	5min	
NOV1F-F	2μL	98°C	30s	
NOV1F-R	2μL	64°C	30s	
Template	20ng	72°C	45s	
DDW	20μL	72°C	5min	
		25°C	∞	



2.Connection

Connecting the two fragments using the Gibson method.

Gibson system (10μL)

C117	5μL
pYB1a-progo	102.3ng
nov1f	29.6ng
DDW	1μL

Gibson

50°C	30min
4°C	∞

3.Transformation

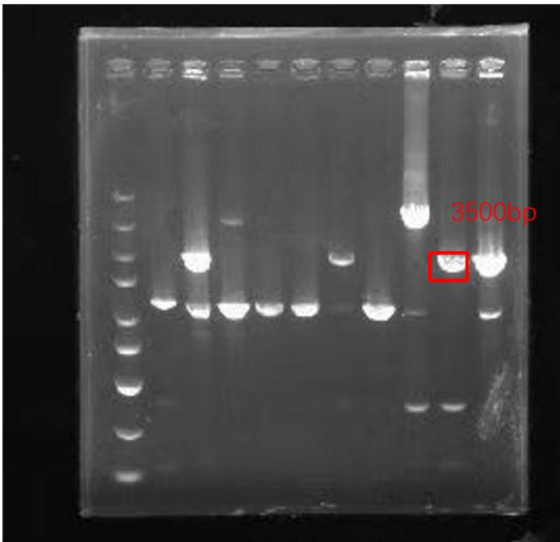
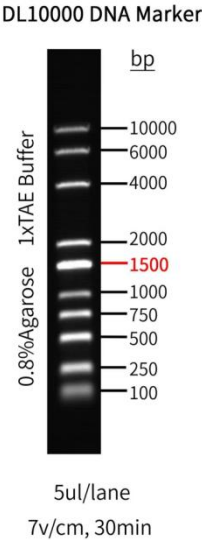
The ligated system was transferred into DH5α and BW25113 receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



4.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 3500bp

PCR system (10μL)		PCR		} ×30
2×Mix	5μL	98°C	5min	
PBF	0.4μL	98°C	30s	
PBR	0.4μL	62°C	30s	
Template	1μL	72°C	1min50s	
DDW	to 10μL	72°C	5min	
		25°C	∞	



(B)Run HPLC to verify pyruvate production

1.Inoculation

Original strains: BW25113、BW-pYB1a-*eGFP*、BW-pYB1a-*vdh*、BW-pYB1a-*progo*、BW-pYB1a-*novIf*、BW-pYB1a-*vdh-nox*

Cultivation in a shaker at 37°C for 12 hours

2.Induction

Induction of BW-pYB1a-*vdh*、BW-pYB1a-*vdh-nox*、BW25113、BW-pYB1a-*progo*、BW-pYB1a-*novIf*、BW-pYB1a-*eGFP*

Induction System Component	Volume
ZY	4.8 mL

50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

BW-pYB1a-*vdh*、BW-pYB1a-*vdh-nox* using vanillin to catalyze the induced enzymes for VA production. BW-pYB1a-*progo* using 4-PG to catalyze the induced enzymes for isoeugenol production. BW-pYB1a-*novIf* using isoeugenol to catalyze the induced enzymes for vanillin production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀ = 6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in pH=7.0 Tris-HCL buffer, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 1、3 hours.

4. High-Performance Liquid Chromatography (HPLC) Detection of VA 、 isoeugenol、 vanillin

1. Preparation of the mobile phase.

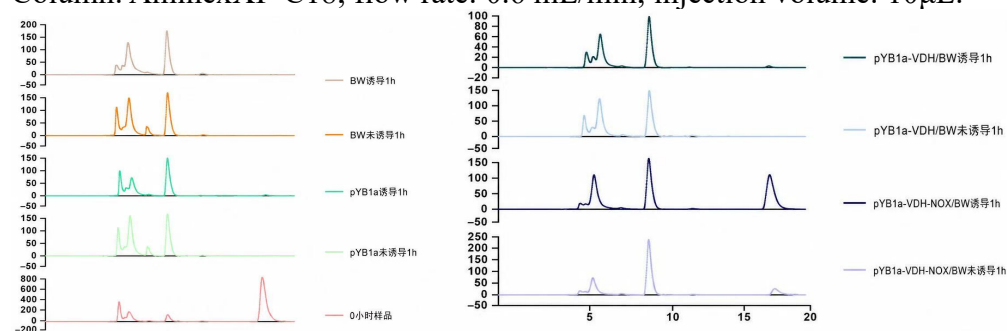
The mobile phase was prepared: methanol and 0.1% glacial acetic acid were also prepared.

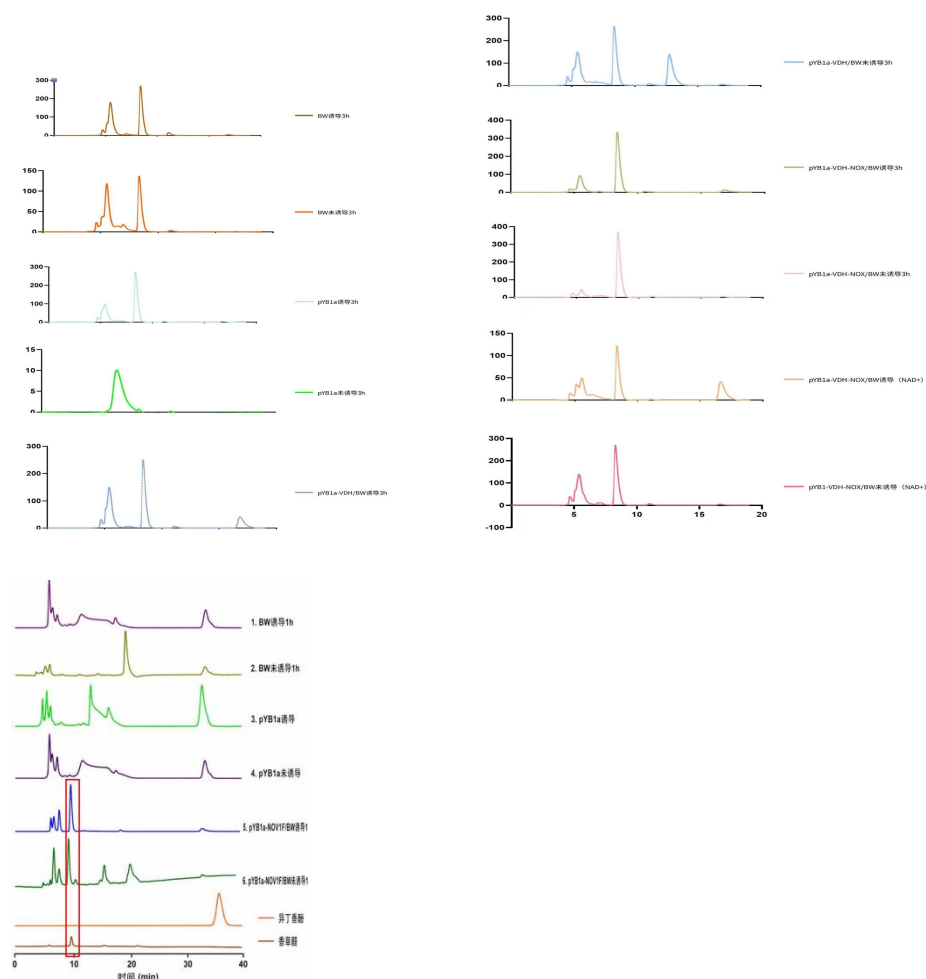
2. Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

3. HPLC detection.

The samples were detected by HPLC using methanol and 0.1% glacial acetic acid . Column: AminexXP-C18; flow rate: 0.6 mL/min; injection volume: 10 μ L.





(B) Validation of single-gene expression by protein analysis

1. Sample Preparation

The pYB1a-Vdh-Nox, pYB1a-Vdh, pYB1a-Progo, pYB1a-Nov1f, pYB1a-eGFPt bacterial strains were cultured under induction conditions at 30°C for 16 hours. The cultures were collected, and the bacterial suspension with a $OD_{600} = 6$ unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

2. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. Transfer the sample into the PCR instrument and heat at 100°C for 10 minutes to denature proteins. After checking for leaks with the buffer solution, mix 32 μ L of prepared sample with 8 μ L loading buffer. After that, load 12 μ L of the mixture into the gel well and add 10 μ L of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

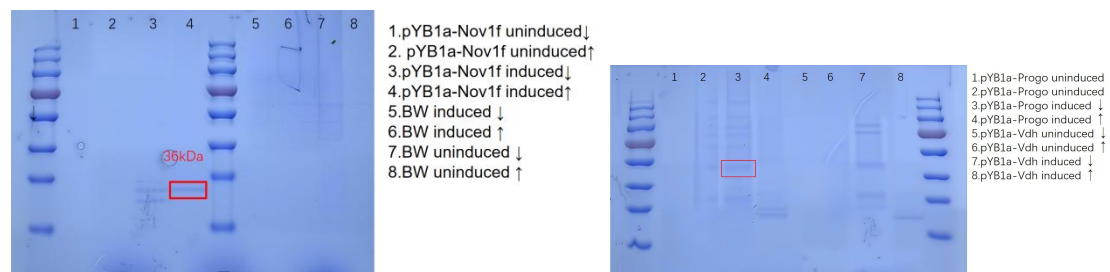
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3.electrophoresis

Connect the power supply and run the gel at 80 V for 90 minutes

4.Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



Although the bands were not prominent owing to insufficient protein loading, the results showed that the protein of pYB1a -Progo was well soluble and could be expressed normally in cells after 30°C induction for 20h.Insufficient protein samples of pYB1a-Nov1f and pYB1a-Vdh may result from inadequate induction time or low OD₆₀₀ of harvested cells.

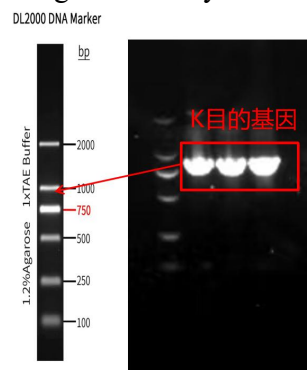
Module 2:

(A) Alteration of plasmid resistance

1.rPCR on pRB1k-ODM.

Gel electrophoresis result detection (6 μL loaded)

Length of kanamycin resistance target gene: 907 bp.



After successful PCR amplification, the PCR product was subjected to enzymatic digestion.

5.Colony PCR

2.Digestion

Digestion system and procedure:

Digestion system (50μL)

PCR system (50μL)

Dpn I	1ul
CutSmart buffer	5ul
PCR product	44ul

three reaction tubes were prepared, and all samples were purified together.

kana resistance target gene fragment
208ng/ μ L

3.Connection

Connecting the two fragments using the Golden-Gate meth

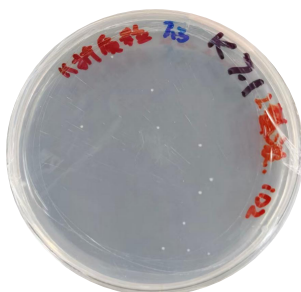
Gibson system (10 μ L)	
C117 enzyme	5 μ L
K-resistant target gene	18.14ng
Template fragment 1	36.42ng
Template fragment 2	63.14ng
DDW	0.53 μ L

Connection	
temperature	time
50°C	30min
25°C	∞

4.Transformation

The ligated system was transferred into DH5 α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;

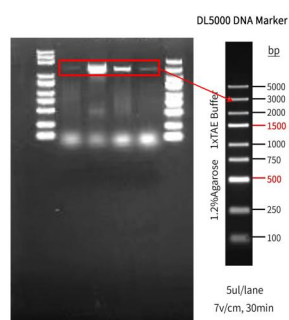
5.Plating results



Four colonies were selected for colony PCR

6.Colony PCR

PBF/PBR was chosen as the primer with a theoretical length of 2758bp.Colony PCR yielded correct target bands, indicating successful ligation.



The colony PCR showed expected band sizes, indicating successful assembly.

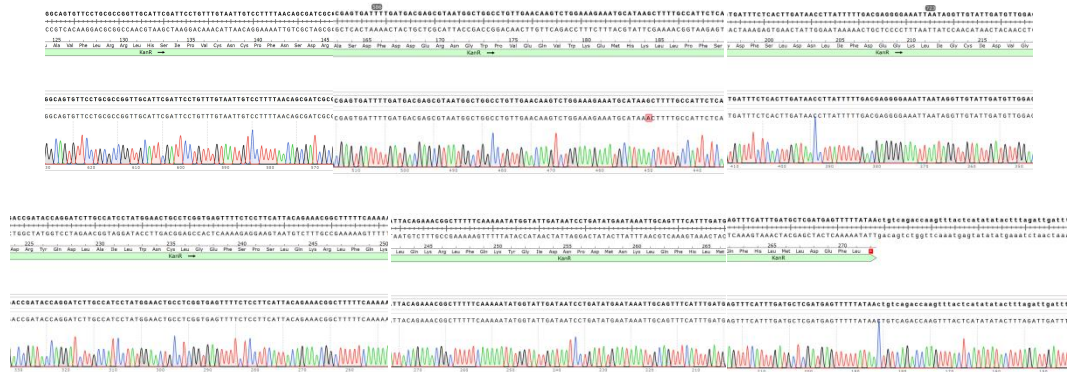
7. Plasmid extraction + send for sequencing

The plasmid was sent for testing.

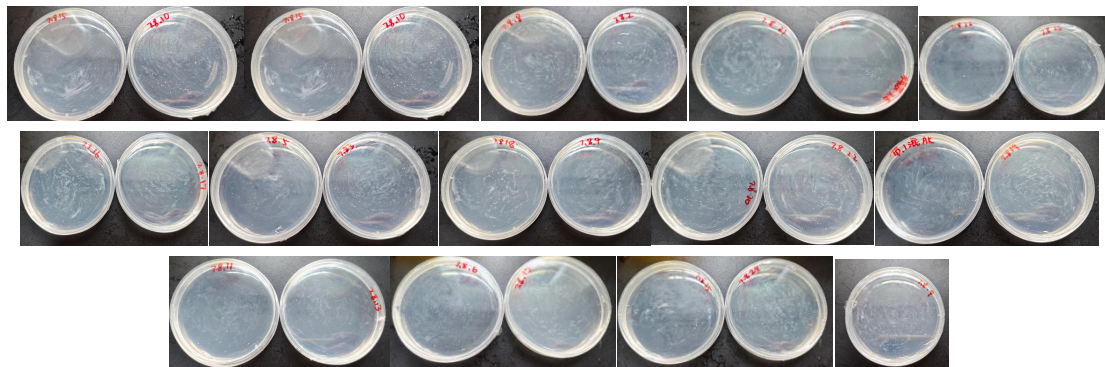
Plasmid extraction results:

Colony 2	Colony 3
327.4ng/μL	369.2ng/μL

Sequencing results confirmed successful assembly.



8. Prepare Competent cells of gRB2k-pobRWT-PpobA *2-mCherry-CD-Cmr.



Single colonies were obtained, yet the library capacity was relatively low, with approximately 300 colonies per plate.

9. Cell scraping

The agar plates with colonies incubated at 37 ° C for 12 h were taken out, and all operations were performed in a biosafety cabinet. Add 5 mL of antibiotic-containing LB liquid medium to each plate, and discard the remaining LB medium.

Use sterile disposable inoculating loops to fully scrape all colonies on each plate to completely suspend the bacterial cells in the medium.

Transfer all bacterial suspensions eluted from each plate into an empty conical flask using a pipette.

Seal the conical flask and incubate it in a shaker at 37 ° C with shaking for 4 – 5 h, followed by strain preservation.

(B) Sample preparation and HPLC detection

1.Induction:

BW, pYB1a-eGFB and pYB1a-TPA bacterial cultures incubated at 37 °C for 12 h were taken. A control group without arabinose induction was set for each strain. Six tubes of ZY medium were transferred into a biosafety cabinet;

The induction system was prepared according to the following formula: 100 µL of 50× M9 stock solution, 100 µL of 5052, 10 µL of magnesium sulfate, 10 µL of 1000× stock, 5 µL of antibiotic, 50 µL of bacterial culture, and 50 µL of 120% arabinose inducer;

Induction was carried out in a shaker at 30 °C for 16 h.

2.Whole-cell catalysis:

The OD value was measured using a spectrophotometer, and the cell concentration was normalized to $OD_{600} = 6$. Calculation formula: Sampling volume (mL) = $6 \div$ measured OD value. The bacterial suspension was aliquoted into 1.5 mL centrifuge tubes according to the calculated volume;

The centrifuge tubes were carefully balanced and centrifuged at 4200 rpm for 10 min. After centrifugation, the supernatant was completely discarded;

The reaction system was prepared as follows: vanillic acid at a final concentration of 500 mg/L (2.98 mM), with 50 mM Tris-HCl buffer (pH 7.5) as the solvent. All components were added to the centrifuge tubes containing cell pellets;

The tubes were incubated in a shaker at 30 °C with shaking at 200 rpm for 12 h under dark conditions.

3.Sample preparation:

The OD value was measured using a spectrophotometer, and the cell concentration was normalized to $OD_{600} = 6$. Calculation formula: Sampling volume (mL) = $6 \div$ measured OD value. The bacterial suspension was aliquoted into 1.5 mL centrifuge tubes according to the calculated volume;

The centrifuge tubes were carefully balanced and centrifuged at 4200 rpm for 10 min. After centrifugation, the supernatant was completely discarded;

The reaction system was prepared as follows: vanillic acid at a final concentration of 500 mg/L (2.98 mM), with 50 mM Tris-HCl buffer (pH 7.5) as the solvent. All components were added to the centrifuge tubes containing cell pellets;

The tubes were incubated in a shaker at 30 °C with shaking at 200 rpm for 12 h under dark conditions.

4.Mobile phase preparation

The mobile phase was prepared at a volume ratio of 0.1% aqueous phosphoric acid solution : methanol = 7 : 3 and mixed thoroughly. 10% methanol and pure methanol were also prepared separately, transferred into large bottles and labelled with name, preparation date and other relevant information.

This system consists of aqueous and organic phases. The aqueous phase was vacuum-filtered into the HPLC solvent reservoir to remove fine impurities.

After filtration, the mobile phase was subjected to sonication for 30 min for degassing to fully eliminate dissolved air bubbles, which prevents unstable baseline and spurious peaks

during HPLC detection.

5.HPLC conditions and results

Mobile phase: 0.1% aqueous phosphoric acid : methanol = 7:3;

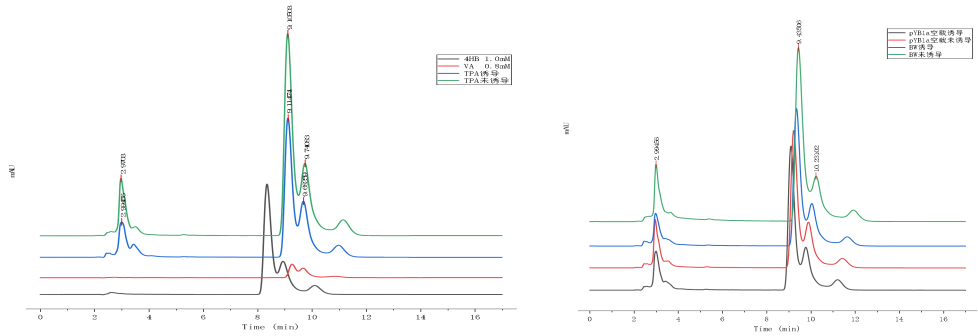
Column: C18, 250 mm × 4.6 mm, 5 μm;

Wavelength: 254 nm;

Injection volume: 10 μL;

Flow rate: 0.9 mL/min;

Run time: 15 min per sample



Module 3:

(A)Run HPLC to verify pyruvate production.

1.Shake culture of strains No.1, 2 and 12, and send for testing

Strains No.1, No.2 and No.12 were inoculated and cultivated in LB medium (37°C, 12 hours), and then submitted for sequencing/testing.

2.Culture preservation and plasmid extraction

The strains were preserved (glycerol stock), and plasmids were extracted and their concentrations were measured.

3.Induction: Induction: Induction of BW-pLB1s-*VanAB-aroY*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μL
50x M supplement	100 μL
MgSO4	10 μL
1000x trace element	10 μL
Bacterial suspension	50 μL
arabinose	50 μL

Antibiotic

5 μ L

cultivation in a shaker at 30°C for 16 hours.

4. Whole-cell catalysis

Measurement of the OD600 value.

BW-pYB1a-dmpB-dmpD-mhpD-mhpE using VA to catalyze the induced enzymes for Pyr production. BW-pLB1s-VanAB-aroY, BW-pYB1a-dmpB-dmpD-mhpD-mhpE using CA to catalyze the induced enzymes for Pyr production. Determine the OD 600 of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD 600=6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in H₂SO₄, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30 ° C for 1 hours.

The whole-cell catalytic system was configured as follows:

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	/	5mM H ₂ SO ₄ solution
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Induced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE))	VA 5mM (final)	/	5mM H ₂ SO ₄ solution
Induced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE))	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Induced BW	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW-pYB1a-eGFP	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Induced BW-pYB1a-eGFP	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	/	5mM H ₂ SO ₄ solution

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	/	5mM H2SO4 solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a-eGFP	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced BW-pYB1a-eGFP	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution

All components were added to the cell pellet tubes and mixed by pipetting, then shaken at 30°C, 200r/min in the dark.

5. Prepare standards and mobile phase

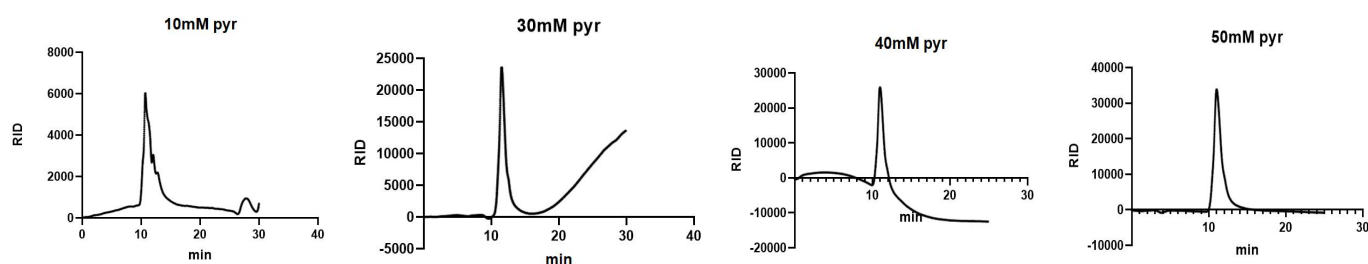
The HPLC standard solutions and the mobile phase (5mM H2SO4 aqueous solution, etc.) were prepared.

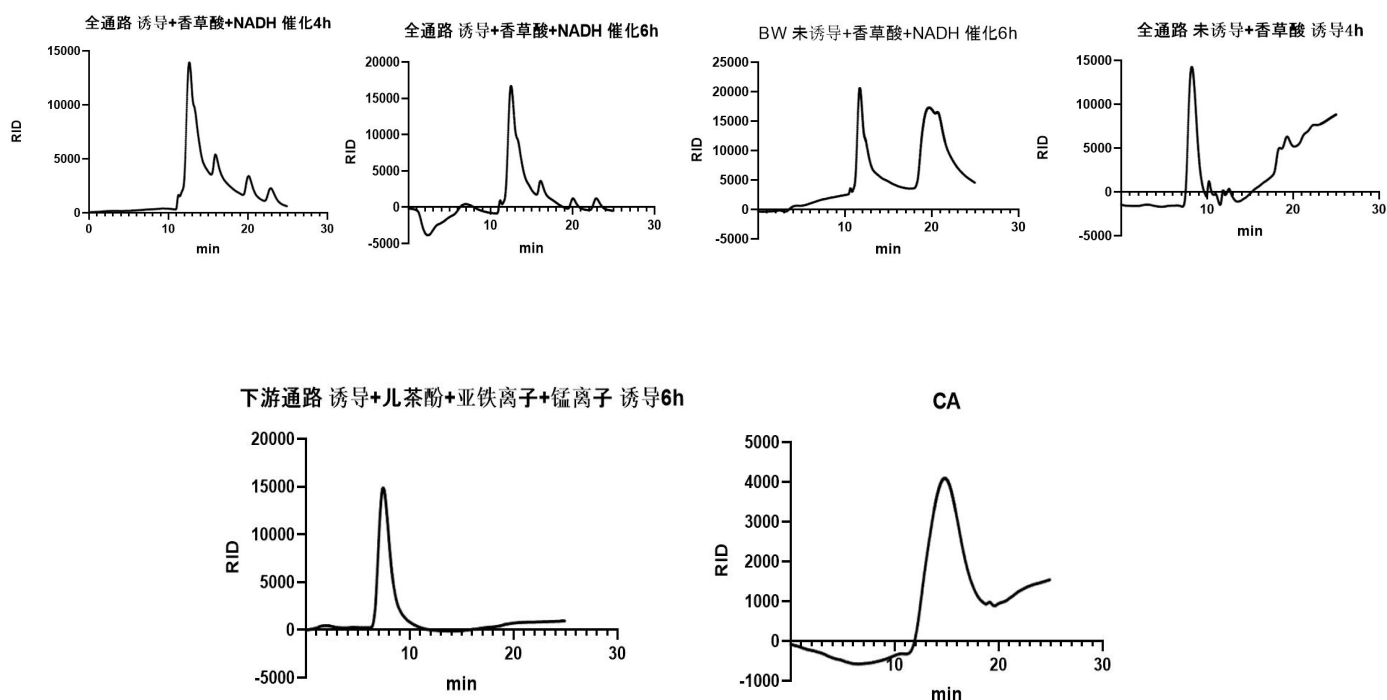
6. HPLC (acetonitrile-ultrapure water method; 5mM dilute H2SO4 sugar-H column method)

The samples were detected by HPLC with the acetonitrile-ultrapure water method (mobile phases: 5% acetonitrile/5mM H2SO4, 30% acetonitrile/5mM H2SO4)

7. HPLC run.

The standards and the samples (full pathway and downstream pathway) were detected by HPLC with RID.





8. Analyze the vanillic acid and pyruvate peaks

The vanillic acid peak appeared at about 14min; in the induced full-pathway group with vanillic acid, a pyruvate peak (tentatively identified) appeared at about 12min. For the downstream-pathway samples, since the RID was extremely sensitive to temperature, an accurate judgment could not be made temporarily.

(B) Validation of single-gene expression by protein analysis

1. Induction.

50μL of bacterial fluid was taken and transferred to the ZY induction system, and cultivated in a shaking incubator at 30°C for 16 hours. The induction system is as follows:

ZY induction system:

Component	Volume
ZY	4.8mL
50×5052 supplement	100μL
50×M supplement	100μL
MgSO ₄	10μL
1000× trace element	10μL
Bacterial suspension	50μL
arabinose	50μL
Antibiotic	5μL

2.Sample Preparation

The cultures were collected, and the bacterial suspension with a $OD_{600}=6$ unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

3.point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. After checking for leaks with the buffer solution, mix 16 μ L of prepared sample with 4 μ L loading buffer. Transfer the mixed sample into the PCR instrument and heat at 100°C for 5 minutes to denature proteins. After cooling, load 16 μ L of the mixture into the gel well and add 4 μ L of protein marker. The sample loading is now complete.

Concentrating Gel:

10% Lower Gel	2 Gel Units	2 Gel Units	4 Gel Units	6 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	20	90	40	30
Water	2.7	5.7	5.5	8.2	8.9	13.2	16.7
30% Acrylamide Solution	4	3.3	6.7	10.0	1.2	13.3	16.7
Tris (pH 8.8)	2.5	1.25	3	3.75	5.0	5.0	6.25
10% SDS	0.1	0.05	0.1	0.15	0.2	0.2	0.25
10% Ammonium	0.1	0.05	0.2	0.15	0.3	0.4	0.25
TEMED	0.002	0.002	0.004	0.006	0.012	0.008	0.01

separation gel:

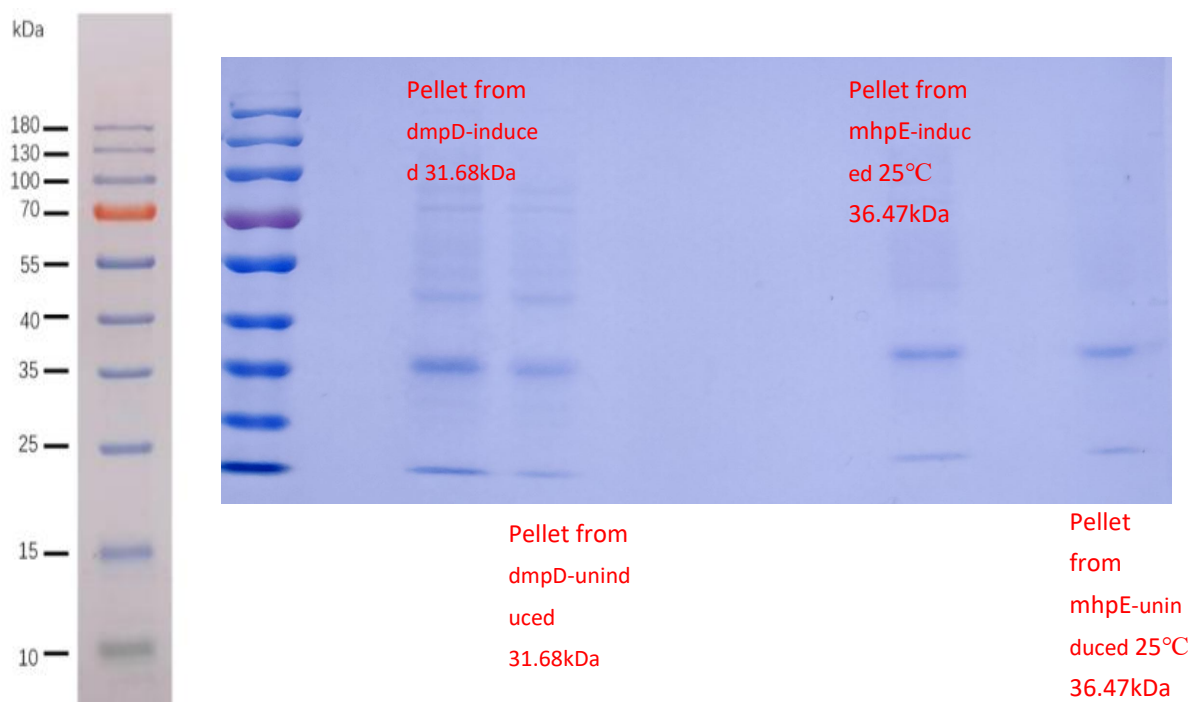
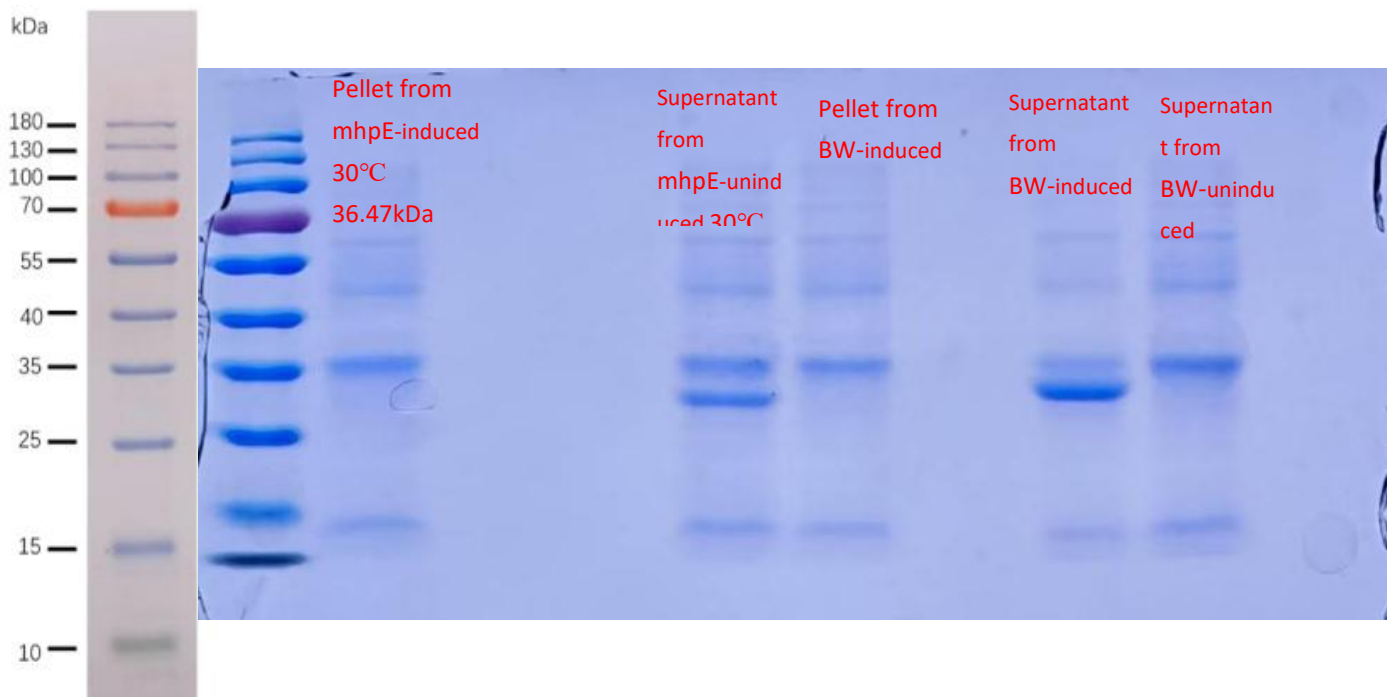
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

4.electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

5.Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:





(C) Characterization of the biosensor response

1. The complete pathway was verified using a pyruvate biosensor.

96-well addition system for full-pathway and downstream-pathway verification. Incubation was performed at 30 ° C for 24h and measure every 2 h.

A-E: the full-pathway strain (pyruvate sensor + pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE);

B-E: the downstream-pathway strain (pyruvate sensor + pYB1a-dmpBD-mhpDE); empty vector: pYB1a-eGFP. Culture volume 2 μ L per well.

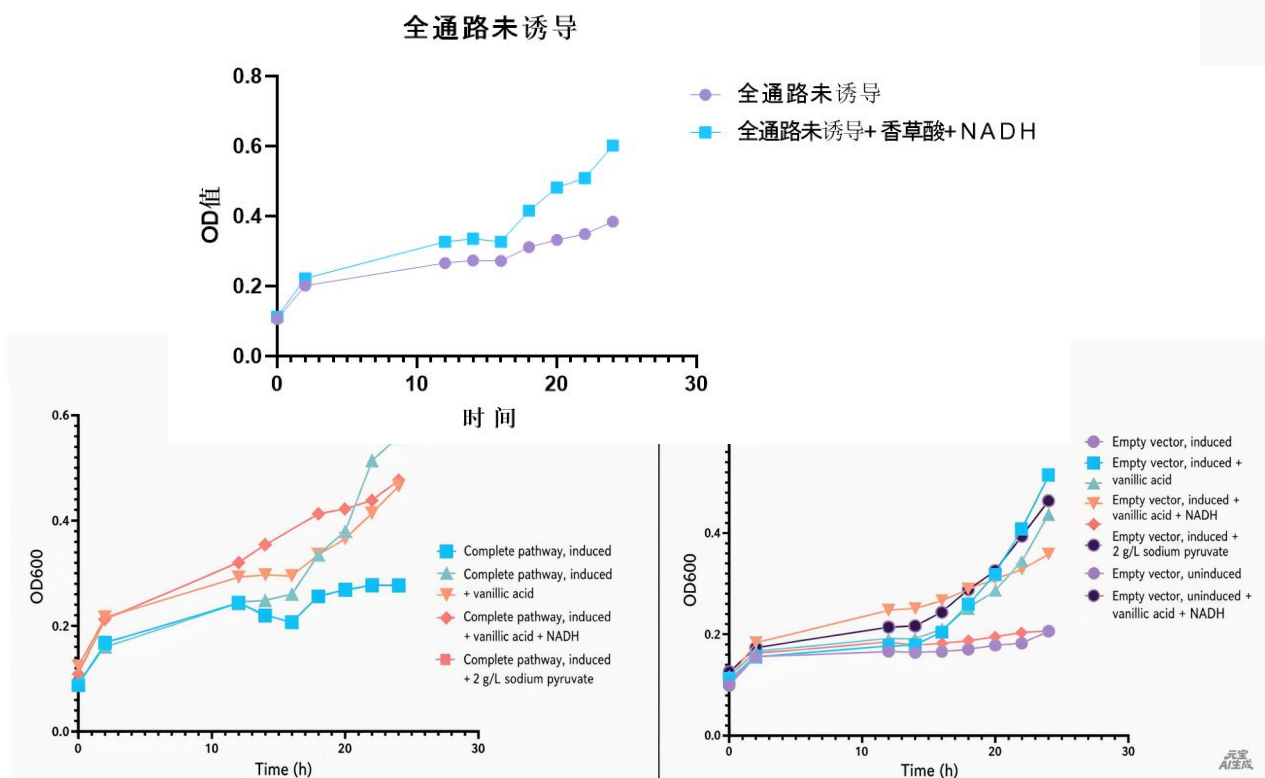
Full-pathway and downstream-pathway groups (per well):

No.	Group	Medium	Additives
1	Induced A-E, blank	198 μ L plain 1 $\times$ M9 (AS)	—
2	Induced A-E + vanillic acid	198 μ L 1 $\times$ M9 with vanillic acid (AS)	—
3	Induced A-E + vanillic acid + NADH	190 μ L 1 $\times$ M9 with vanillic acid (AS)	8 μ L NADH stock
4	Induced A-E + sodium pyruvate (positive)	158 μ L plain 1 $\times$ M9 (AS)	40 μ L NaPyr stock
5	Uninduced A-E, blank	198 μ L plain 1 $\times$ M9 (AS)	—
6	Uninduced A-E + vanillic acid + NADH	190 μ L 1 $\times$ M9 with vanillic acid (AS)	8 μ L NADH stock

No.	Group	Medium	Additives
7	Induced eGFP, blank	198μL plain 1×M9 (A)	—
8	Induced eGFP + vanillic acid	198μL 1×M9 with vanillic acid (A)	—
9	Induced eGFP + vanillic acid + NADH	190μL 1×M9 with vanillic acid (A)	8μL NADH stock
10	Induced eGFP + sodium pyruvate (positive)	158μL plain 1×M9 (A)	40μL NaPyr stock
11	Induced B-E, blank	198μL plain 1×M9 (A)	—
12	Induced B-E + catechol	188μL plain 1×M9 (A)	10μL catechol
13	Induced B-E + catechol + Fe ²⁺ /Mn ²⁺	185μL plain 1×M9 (A)	10μL catechol, 2μL Mn ²⁺ , 1μL Fe ²⁺ (added last)
14	Induced B-E + sodium pyruvate (positive)	158μL plain 1×M9 (A)	40μL NaPyr stock
15	Uninduced B-E, blank	198μL plain 1×M9 (A)	—
16	Uninduced B-E + catechol	188μL plain 1×M9 (A)	10μL catechol
17	Induced eGFP + catechol	188μL plain 1×M9 (A)	10μL catechol
18	Induced eGFP + catechol + Fe ²⁺ /Mn ²⁺	185μL plain 1×M9 (A)	10μL catechol, 2μL Mn ²⁺ , 1μL Fe ²⁺ (added last)

2. Distilled water containing 1 mM vanillic acid was prepared and adjusted to pH ≈ 7.1 with NaOH.

The left panel shows that the addition of vanillic acid markedly promoted the growth of the pyruvate-deficient biosensor, suggesting that the complete pathway is functional. However, the right panel (empty-vector control) reveals that vanillic acid and NADH both benefited growth. Combined with the previous results, we now suspect that sodium ions may severely affect the growth rate, which remains to be verified.



3. Downstream pathway verification (Fe²⁺ interference)

The addition of Fe²⁺ gave the medium a color, which made it impossible to detect cell growth with the microplate reader; the addition of catechol alone had little effect on the growth rate, so the effectiveness of the downstream pathway could not be determined.

Week 8 (7.10-7.16)

Module 1:

(A)Verification of BW-pYB1a-*progo-novIf*.

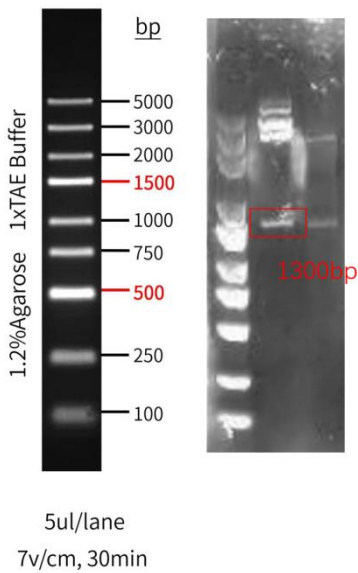
1.Enzyme digestion

Selected 10 colonies for using XhoI BglIII to verify the BW-pYB1a-*progo-novIf*.
Enzyme digestion system:

Enzyme digestion (10μL)	
XhoI	0.2μL
BglIII	0.2μL
rCutsmart	1.0μL
Plasmid	200ng
DDW	to 10μL

The products were analyzed by agarose gel electrophoresis. The results showed that the enzyme digestion was successful as follows:

DL5000 DNA Marker



(B)Run HPLC to verify VA、vanillin production.

1. Inoculation

Original strains: BW25113、BW-pYB1a-*eGFP*、BW-pYB1a-*vdh*、BW-pYB1a-*vdh-nox*、
BW-pYB1a-*progo*、 BW-pYB1a-*novIf*
Cultivation in a shaker at 37°C for 12 hours

2.Induction:

Induction of BW-pYB1a-*vdh*、BW-pYB1a-*vdh-nox*、BW25113、BW-pYB1a-*progo*、BW-pYB1a-*novIf*、BW-pYB1a-*eGFP*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis

BW-pYB1a-*vdh* and BW-pYB1a-*vdh-nox* using vanillin to catalyze the induced enzymes for VA production. BW-pYB1a-*novIf* using isoeugenol to catalyze the induced enzymes for vanillin production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀ = 6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, BW-pYB1a-*vdh*、BW-pYB1a-*vdh-nox* resuspend the bacterial cells in pH=7.0 Tris-HCL buffer, BW-pYB1a-*novIf* resuspend the bacterial cells in pH=7.5 Tris-HCL buffer. Then transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 1、3 hours.

4. High-Performance Liquid Chromatography (HPLC) Detection of VA 、vanillin

1.Preparation of the mobile phase.

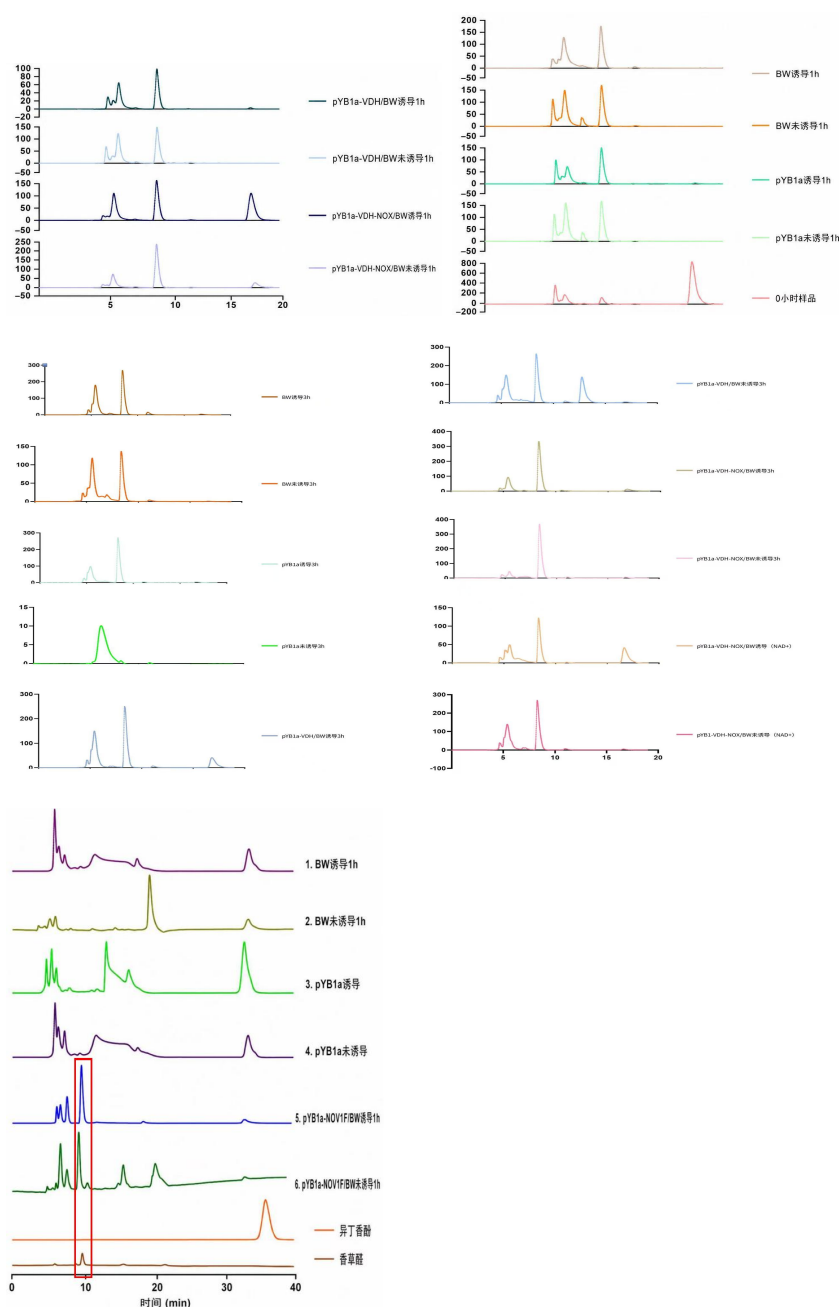
The mobile phase was prepared: methanol and 0.1% glacial acetic acid were also prepared.

2.Preparation of standards and sample filtration.

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

3.HPLC detection.

The samples were detected by HPLC using the methanol and 0.1% glacial acetic acid method. Column: Aminex XP-C18; flow rate: 0.5 mL/min; injection volume: 10 μ L.



(B) Validation of single-gene expression by protein analysis

1. Sample Preparation

The pYB1a-Vdh-NOX, pYB1a-Vdh, pYB1a-Progo, pYB1a-Nov1f bacterial strains were cultured under induction conditions at 30°C for 16 hours. The cultures were collected, and the bacterial suspension with a $OD_{600} = 6$ unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

2.point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. Transfer the sample into the PCR instrument and heat at 100°C°C for 10 minutes to denature proteins. After checking for leaks with the buffer solution, mix 32 µL of prepared sample with 8 µL loading buffer. After that, load 12 µL of the mixture into the gel well and add 10 µL of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

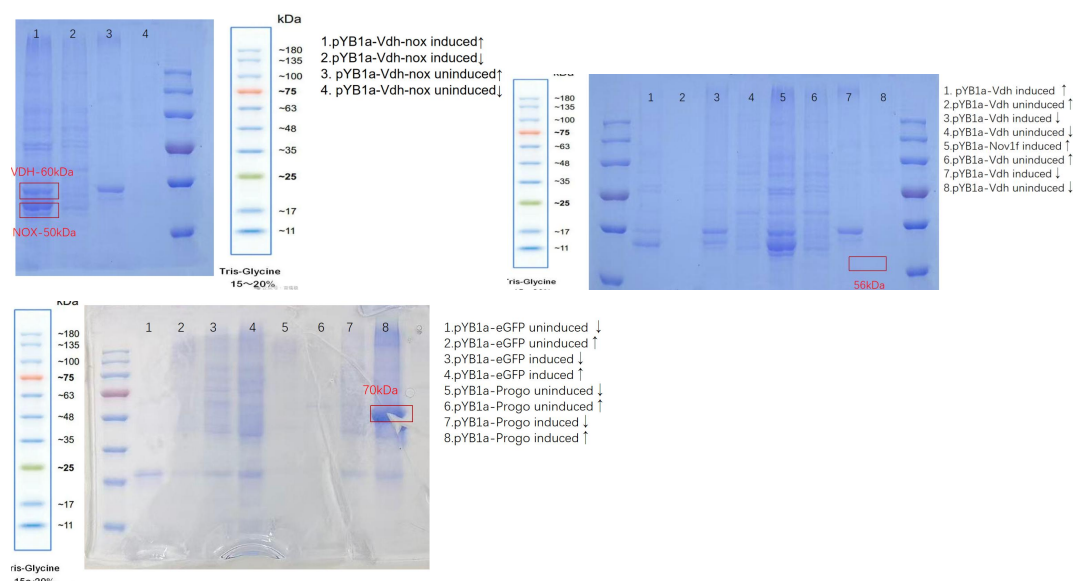
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3.electrophoresis

Connect the power supply and run the gel at 80 V for 90 minutes

4.Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



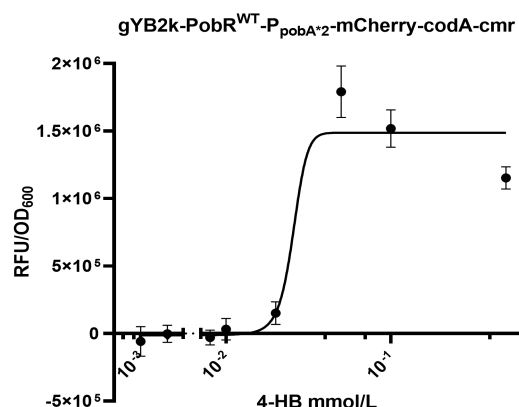
Although the bands were not prominent owing to insufficient protein loading, the results showed that the protein of pYB1a -Progo was well soluble and could be expressed normally in cells after 30°C induction for 20h.

Module 2:

(A) Characterization of the biosensor response

1. Determination of biosensor response curve to 4-HB.

Determination of the biosensor response curve to 4-HB. The reaction system was set to a total volume of 200 μ L, containing 1 μ L bacterial culture, 0.2 μ L kanamycin stock, 4-HB and 1 $\times$ M9 medium. Incubation was performed at 37 ° C for 16 h. The concentration gradient of 4-HB was set as 0, 0.001, 0.002, 0.008, 0.01, 0.02, 0.05, 0.1 and 0.5 mM. Six replicates were prepared for each 4-HB concentration.

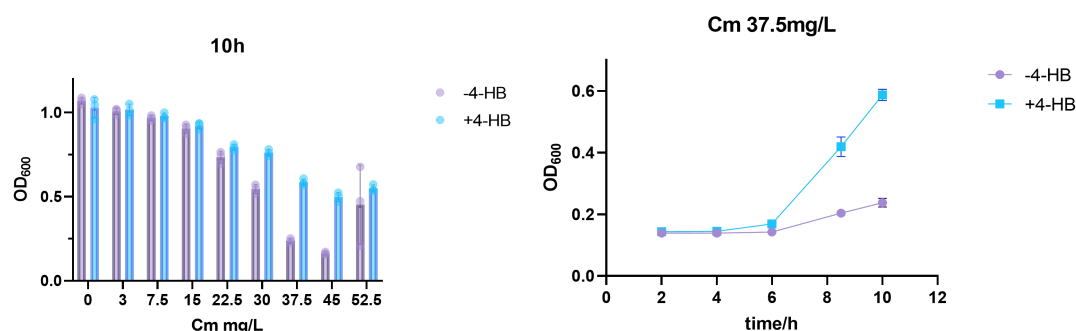


After excluding the risk of pipette tip contamination, the measured values still decreased as the 4-HB concentration increased from 0.1 to 0.5 mM, which was consistent with the first batch of tests. This phenomenon may be caused by insufficient mixing of the premixed system or unstable sample loading volume.

2. Determination of saturation concentration of biosensor response to Cm

200 μ L system: 1 μ L bacterial culture + 0.2 μ L kanamycin + 4-HB + LB + chloramphenicol. The optimal stable 4-HB concentration was 0.6 mmol/L. A control without 4-HB was set. Incubate at 37 ° C for 10 h, and measure every 2 h.

Chloramphenicol gradient: 0, 3, 7.5, 15, 22.5, 30, 37.5, 45, 52.5 mg/L. Three replicates for -4-HB and +4-HB at each concentration.



Optimal chloramphenicol concentration was 37.5 mg/L, matching expectations. Nevertheless, the difference between +4HB and -4HB controls was not as prominent as before antibiotic replacement. It is speculated that high-concentration 4HB caused inhibitory effects after the antibiotic switch.

(B)Sample preparation and HPLC detection

1.Induction:

BW, pYB1a-*eGFP* and pYB1a-*odm* strains incubated at 37 ° C for 12 h were taken out. A non-ara induction control was set for each group, and six tubes of ZY medium were transferred into the biosafety cabinet.

The induction system was prepared according to the following formula: 100 μ L of 50 $\times$ M stock, 100 μ L of 5052 solution, 10 μ L magnesium sulfate, 10 μ L of 1000 $\times$ stock, 5 μ L antibiotic, 50 μ L bacterial culture and 50 μ L of 20% arabinose inducer.

Induction was performed in a shaker at 30 ° C for 16 h.

2.Whole-cell catalysis:

The OD values were determined by spectrophotometry, and the cell density was normalized to OD₆₀₀ = 6. The sampling volume (mL) was calculated as: Sampling volume = 6 / measured OD value. The bacterial suspension was aliquoted into 1.5 mL centrifuge tubes according to the calculated volume.

Centrifuge tubes were carefully balanced and centrifuged at 4200 rpm for 10 min. After centrifugation, the tubes were taken out and all supernatants were discarded.

The reaction system was prepared with vanillic acid at a final concentration of 10 mM in 50 mM Tris-HCl buffer (pH 7.5). The prepared solution was added to the cell pellets in each centrifuge tube.

One 0 h control sample was prepared for each group (12 samples in total) and stored at -20 °C. The catalytic

reaction groups were incubated in a shaker at 30 °C with shaking at 200 r/min under dark conditions for 12 h.

3.Sample preparation:

All samples after catalytic reaction were collected and centrifuged at 13000 rpm for 10 min to fully pellet bacterial cells.

The clear supernatant was carefully aspirated.

The supernatant was filtered through a 0.22 µm aqueous-phase membrane filter to remove impurities, and the filtrate was collected in a collection vial.

The filtrate was transferred into new HPLC inserts, labelled with sample information, and placed into reusable HPLC sample vials.

All prepared samples were stored at -20 °C in a refrigerator before HPLC analysis.

4.Mobile phase preparation

The mobile phase was prepared at a volume ratio of 0.1% aqueous phosphoric acid solution : methanol = 75 : 25 and mixed thoroughly. 10% methanol and pure methanol were prepared separately, transferred into large bottles and labelled with medium name, preparation date and other relevant information.

This mobile phase consists of aqueous and organic components. The aqueous fraction was vacuum-filtered into the HPLC solvent reservoir to remove fine particles.

After vacuum filtration, the mobile phase was subjected to sonication for 30 min for degassing, so as to fully eliminate dissolved air bubbles and prevent baseline drift and spurious peaks during chromatographic detection.

5.HPLC conditions and results

Mobile phase: 0.1% aqueous phosphoric acid : methanol = 75:25;

Column: C18, 250 mm × 4.6 mm, 5 µm;

Wavelength: 254 nm;

Injection volume: 10 µL;

Flow rate: 0.9 mL/min;

Run time: 15 min per sample

(C) Preliminary screening assay

1.System

The M9 screening system was prepared to test different concentrations of bacterial culture, vanillic acid and chloramphenicol.

VA concentration: 2.5 mM (prepared from 10 mM stock solution; the pH of diluted VA solution was 3.3, which was adjusted to pH 6.65 with NaOH).(2) Bacterial culture volumes tested: 5 µL, 50 µL and 250 µL.(3)

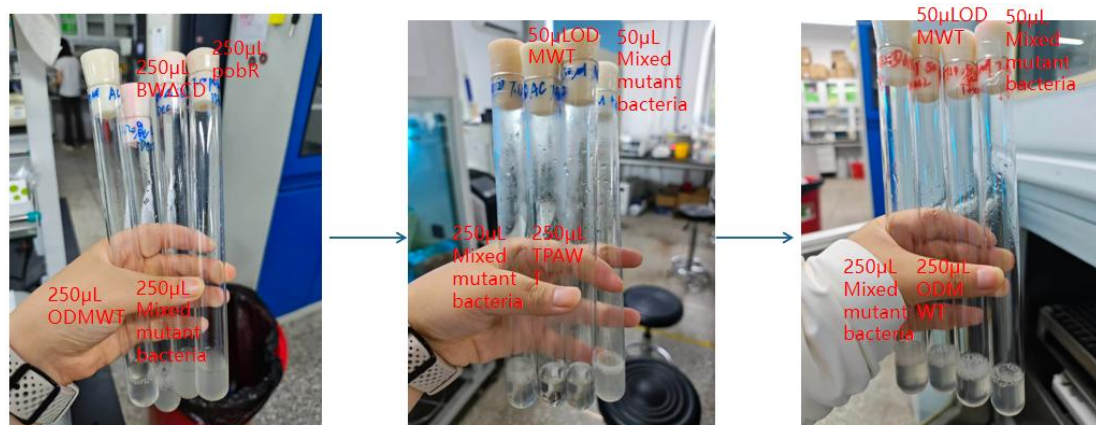
Chloramphenicol concentration: 30 mg/L

Detailed reaction system (5mL) :

Components	WT	pobR	BWΔCD	mix
Antibiotics	A 5μL	K 5μL	无	A 5μL, K 5μL
5×M9	1 mL	1 mL	1 mL	1 mL
20% glycerol	500 μL	500 μL	500 μL	500 μL
MgSO ₄	10 μL	10 μL	10 μL	10 μL
CaCl ₂	0.5 μL	0.5 μL	0.5 μL	0.5 μL
Cm	5 μL	5 μL	5 μL	5 μL
Water + VA + bacterial culture	3479.5 μL + 5 μL bacterial culture	3479.5 μL + 5 μL bacterial culture	3479.5 μL + 5 μL bacterial culture	3469.5 μL + 5 μL bacterial culture
Water + VA + bacterial culture	3429.5 μL + 50 μL bacterial culture	3429.5 μL + 50 μL bacterial culture	3434.5 μL + 50 μL bacterial culture	3424.5 μL + 50 μL bacterial culture
Water + VA + bacterial culture	3229.5 μL + 250 μL bacterial culture	3229.5 μL + 250 μL bacterial culture	3234.5 μL + 250 μL bacterial culture	3224.5 μL + 250 μL bacterial culture

2.Screening results

The 5 μL bacterial culture system was abandoned due to slow growth (>20 h) and no obvious distinction. Screening results for 50 μL and 250 μL bacterial culture groups are as follows.

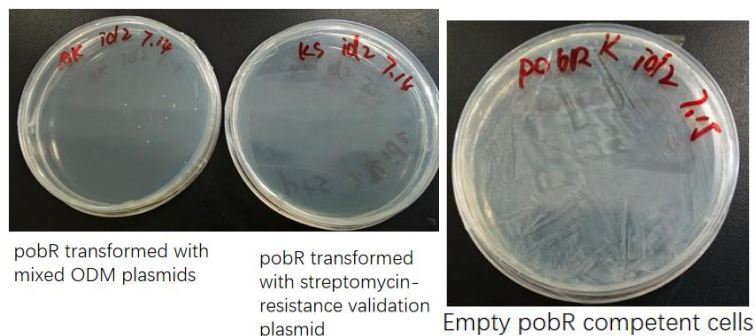


250 μL bacterial culture system: The first transfer was conducted after 12–16 h incubation, and the second transfer after 8–10 h incubation. No obvious difference after the second transfer, presumably due to overlong incubation time.

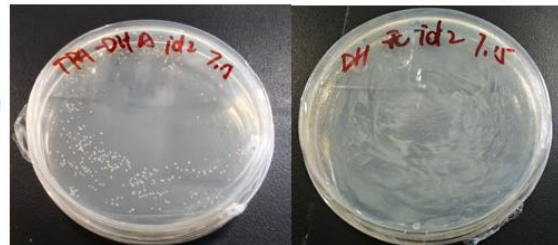
50 μL bacterial culture system: The first transfer was finished after 16–20 h incubation. No obvious difference was observed at ~13 h in the first transfer group, which was considered to result from excessive incubation time.

3.Competent cell validation results

Final OD measurement: 0.418



DH5α transformed with mixed TPA plasmid



Empty DH5α competent cells

3.Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;

Mixed plasmid concentration: 30 ng/μL. Add 1 μL mixed plasmid to competent cells after 5 min on ice. Incubate on ice for 20 min, heat shock at 42 °C for 45 s, then ice bath for 2–3 min. Add 600 μL LB into EP tube, shake at 37 °C for 45 min. Centrifuge at 4200 r/min for 2 min. Take 100 μL aliquot in clean bench, mix and spread on plates.

Module 3:

(A)Experimental design and 96-well plate variable setting

1.Determination of biosensor response curve to PyrNA.

Determination of the biosensor response curve to PyrNA. The reaction system was set to a total volume of 200 μL, containing 1μL bacterial culture, 0.2 μL ampicillin + streptomycin, PyrNA and 1 × M9 medium. Incubation was performed at 37 °C for 24h and measure every 2 h. The concentration gradient of PyrNA was set as 0, 0.5 g/L, 1 g/L, 1.5 g/L, 2 g/L, 2.5 g/L, 3 g/L and BW eight replicates were prepared for each PyrNA concentration.

2.96-well addition system for full-pathway and downstream-pathway verification.

A-E: the full-pathway strain (pyruvate sensor + pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE); B-E: the downstream-pathway strain (pyruvate sensor + pYB1a-dmpBD-mhpDE); empty vector: pYB1a-eGFP. Culture volume 2μL per well.

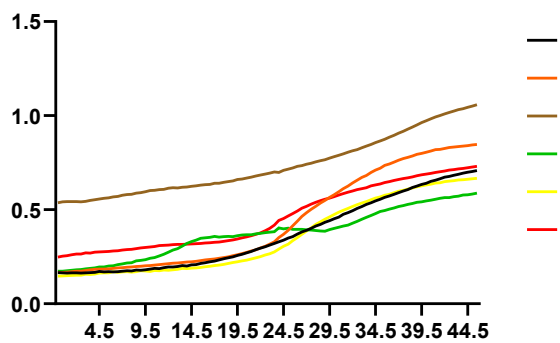
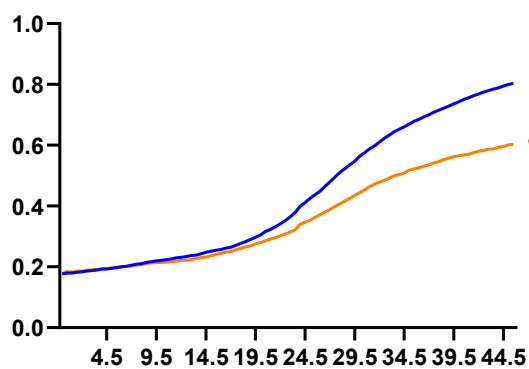
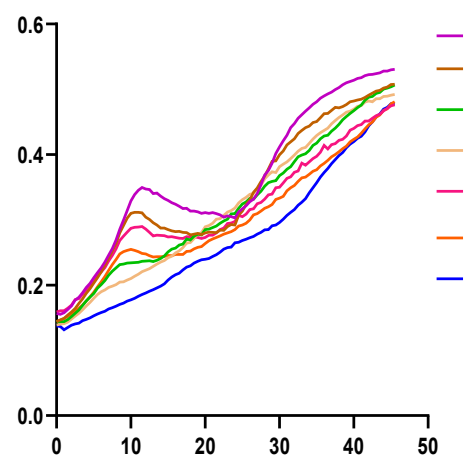
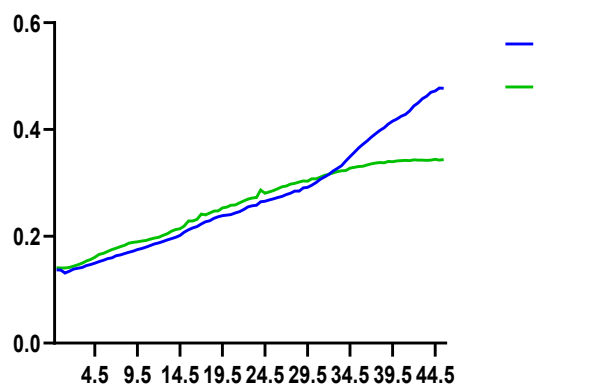
Full-pathway and downstream-pathway groups (per well):

No.	Group	Medium	Additives
1	Induced A-E, blank	198μL plain 1×M9 (AS)	—
2	Induced A-E + vanillic acid	198μL 1×M9 with vanillic acid (AS)	—
3	Induced A-E + vanillic acid + NADH	190μL 1×M9 with vanillic acid (AS)	8μL NADH stock

No.	Group	Medium	Additives
4	Induced A-E + sodium pyruvate (positive)	158μL plain 1×M9 (AS)	40μL NaPyr stock
5	Uninduced A-E, blank	198μL plain 1×M9 (AS)	—
6	Uninduced A-E + vanillic acid + NADH	190μL 1×M9 with vanillic acid (AS)	8μL NADH stock
7	Induced eGFP, blank	198μL plain 1×M9 (A)	—
8	Induced eGFP + vanillic acid	198μL 1×M9 with vanillic acid (A)	—
9	Induced eGFP + vanillic acid + NADH	190μL 1×M9 with vanillic acid (A)	8μL NADH stock
10	Induced eGFP + sodium pyruvate (positive)	158μL plain 1×M9 (A)	40μL NaPyr stock
11	Induced B-E, blank	198μL plain 1×M9 (A)	—
12	Induced B-E + catechol	188μL plain 1×M9 (A)	10μL catechol
13	Induced B-E + catechol + Fe ²⁺ /Mn ²⁺	185μL plain 1×M9 (A)	10μL catechol, 2μL Mn ²⁺ , 1μL Fe ²⁺ (added last)
14	Induced B-E + sodium pyruvate (positive)	158μL plain 1×M9 (A)	40μL NaPyr stock
15	Uninduced B-E, blank	198μL plain 1×M9 (A)	—
16	Uninduced B-E + catechol	188μL plain 1×M9 (A)	10μL catechol
17	Induced eGFP + catechol	188μL plain 1×M9 (A)	10μL catechol
18	Induced eGFP + catechol + Fe ²⁺ /Mn ²⁺	185μL plain 1×M9 (A)	10μL catechol, 2μL Mn ²⁺ , 1μL Fe ²⁺ (added last)

4. Microplate reader program setting.

The program was set as follows: temperature control at 30°C (fluctuation range 29.5-30.5°C); orbital shaking, amplitude 2.5mm, 216rpm, 4s per cycle; total running duration 23h59min59s, one round of detection every 30min; absorbance at 600nm, bandwidth 3.5, 10 flashes, 50ms settling time before reading.



(B)Run HPLC to verify pyruvate production.

1.Shake culture of strains No.1, 2 and 12, and send for testing

Strains No.1, No.2 and No.12 were inoculated and cultivated in LB medium (37°C, 12 hours), and then submitted for sequencing/testing.

2.Culture preservation and plasmid extraction

The strains were preserved (glycerol stock), and plasmids were extracted and their concentrations were measured.

3.Induction: Induction: Induction of BW-pLB1s-*VanAB-aroY*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 µL
50x M supplement	100 µL
MgSO ₄	10 µL
1000x trace element	10 µL
Bacterial suspension	50 µL
arabinose	50 µL
Antibiotic	5 µL

cultivation in a shaker at 30°C for 16 hours.

4.Whole-cell catalysis

Measurement of the OD600 value.

BW-pYB1a-dmpB-dmpD-mhpD-mhpE using VA to catalyze the induced enzymes for Pyr production. BW-pLB1s-VanAB-aroY, BW-pYB1a-dmpB-dmpD-mhpD-mhpE using CA to catalyze the induced enzymes for Pyr production . Determine the OD 600 of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD 600=6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in H₂SO₄, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30 ° C for 1 hours.

The whole-cell catalytic system was configured as follows:

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	/	5mM H ₂ SO ₄ solution

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE)	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE))	VA 5mM (final)	/	5mM H2SO4 solution
Induced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE))	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced BW	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced BW	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a- <i>eGFP</i>	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced BW-pYB1a- <i>eGFP</i>	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	/	5mM H2SO4 solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	/	5mM H2SO4 solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a- <i>eGFP</i>	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Induced BW-pYB1a- <i>eGFP</i>	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution

All components were added to the cell pellet tubes and mixed by pipetting, then shaken at 30°C, 200r/min in the dark.

5.Prepare standards and mobile phase

The HPLC standard solutions and the mobile phase (5mM H₂SO₄ aqueous solution, etc.) were prepared.

6.HPLC with the sugar-H column: standards, VA, CA, full-pathway and downstream-pathway samples

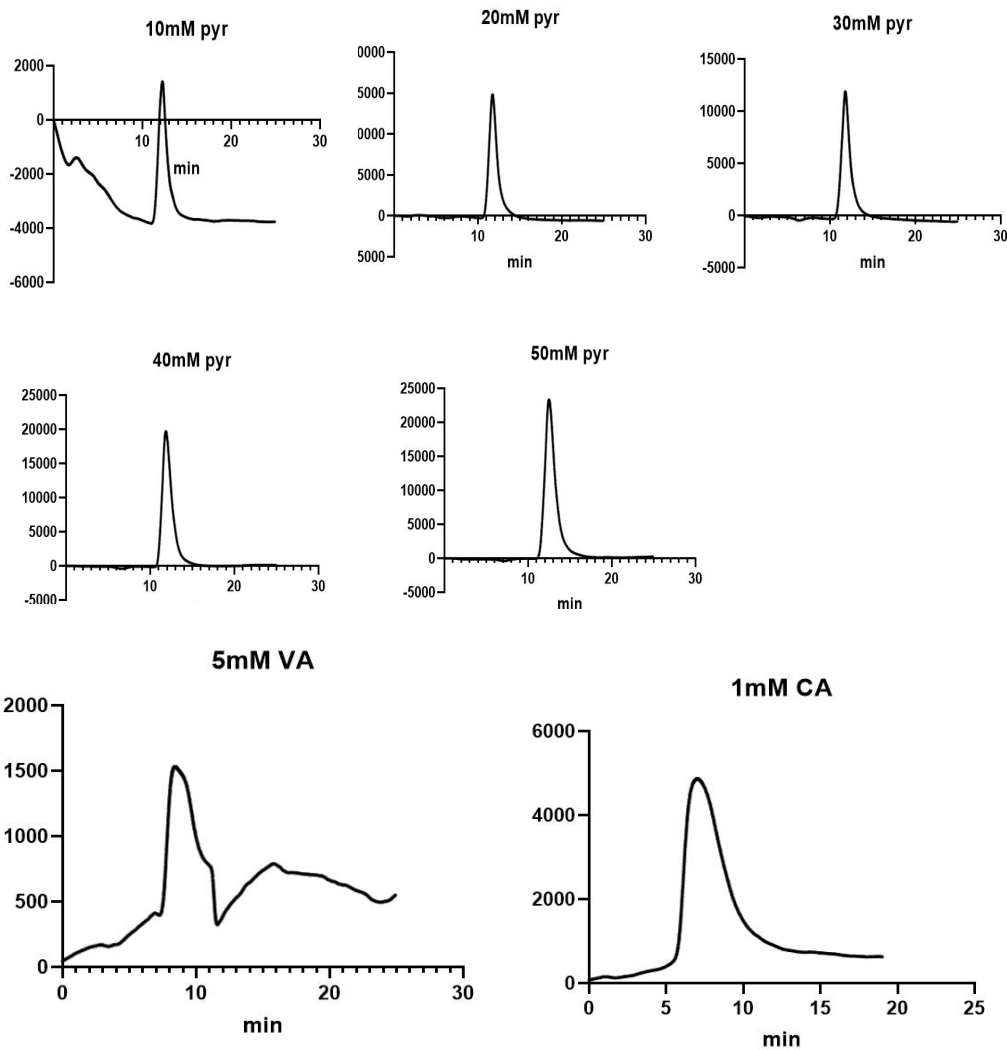
Mobile phase: 5mM H₂SO₄ aqueous solution; flow rate: 0.2 mL/min; column temperature: 50°C; column: sugar-H column; injection volume: 5μL (changed from 10μL according to the manual).

The pyruvate standard peaked at about 12min, with the peak height increasing with concentration; the vanillic acid (VA) and catechol (CA) standards peaked at about 8min and 7min, respectively, both earlier than the pyruvate peak.

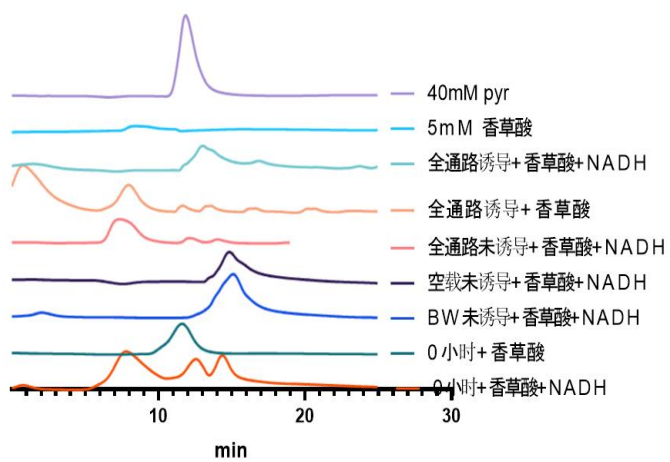
The full-pathway (4h, 12h) and downstream-pathway (4h, 12h) samples were detected by HPLC.

7.HPLC run.

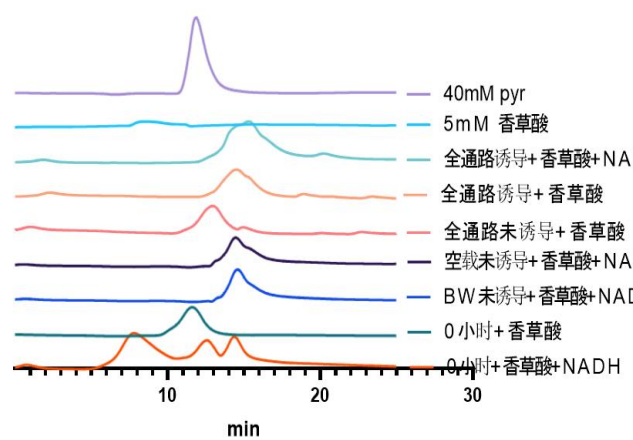
The standards and the samples (full pathway and downstream pathway) were detected by HPLC with RID.

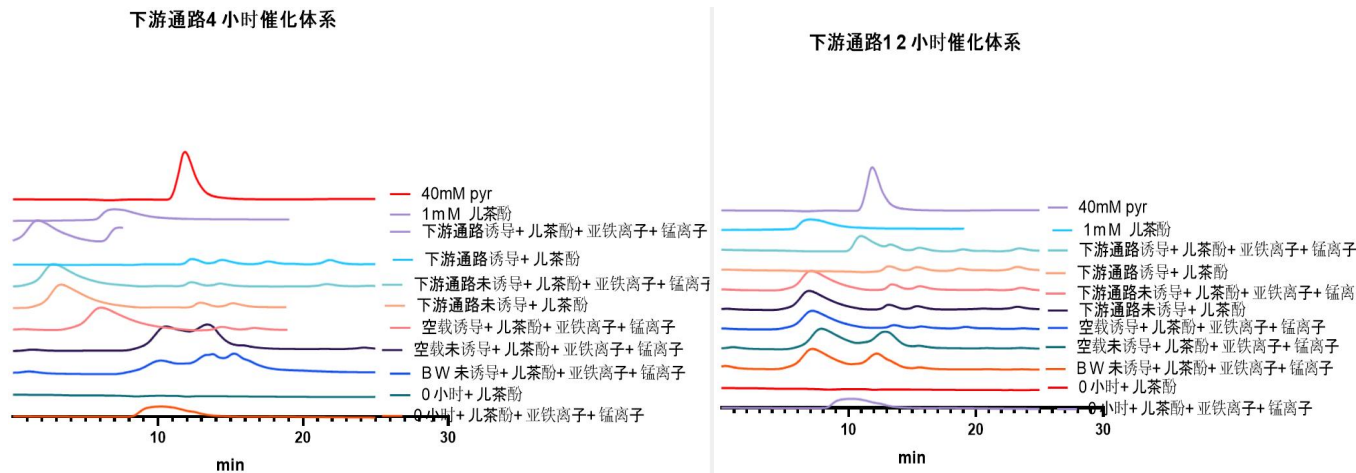


全通路-4小时催化体系



全通路12小时催化体系





Week 9 (7.17-7.23)

Module 1:

(A)Run HPLC to verify pyruvate production.

1. Inoculation

Original strains: BW25113、BW-pYB1a-*eGFP*、BW-pYB1a-*vdh*、BW-pYB1a-*progo*、BW-pYB1a-*progo-novIf*、BW-pYB1a-*vdh-nox*

Cultivation in a shaker at 37°C for 12 hours

2.Induction

Induction of BW-pYB1a-*vdh*、BW25113、BW-pYB1a-*progo*、BW-pYB1a-*eGFP*、BW-pYB1a-*progo-novIf*、BW-pYB1a-*vdh-nox*

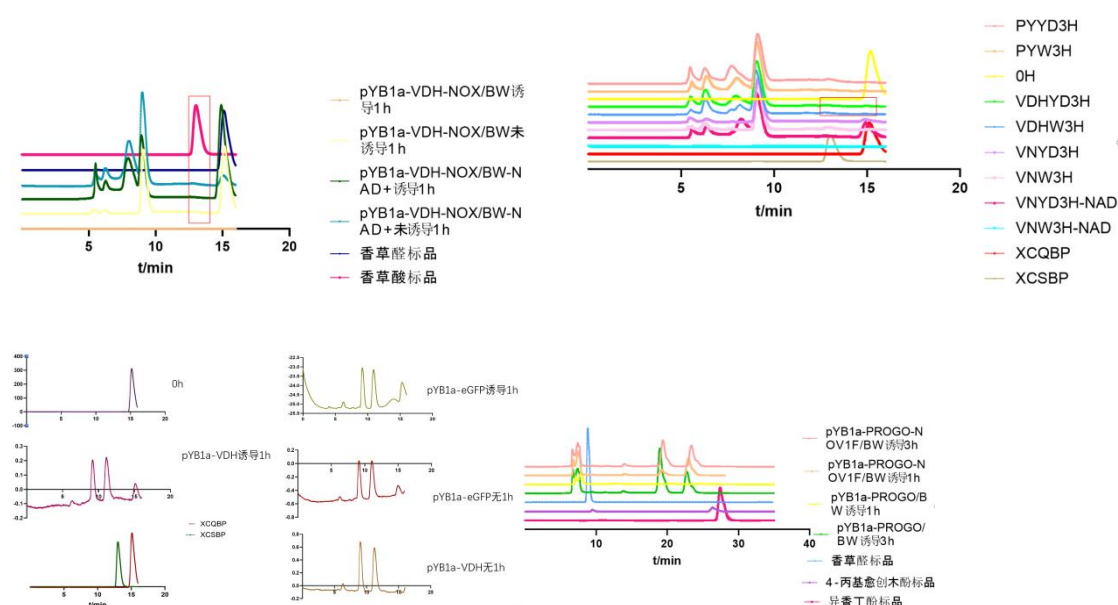
Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis

BW-pYB1a-*vdh*, BW-pYB1a-*vdh-nox* using vanillin to catalyze the induced enzymes for VA production. BW-pYB1a-*progo* using 4-PG to catalyze the induced enzymes for isoeugenol production. BW-pYB1a-*progo-nov1f* using 4-PG to catalyze the induced enzymes for vanillin production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀ = 6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in pH=7.0 Tris-HCL buffer, BW-pYB1a-*progo* resuspend the bacterial cells in pH=7.5 KPi buffer. Then transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 1、3 hours.

4. High-Performance Liquid Chromatography (HPLC) Detection of VA、vanillin、isoeugenol



(B) Validation of single-gene expression by protein analysis

1. Sample Preparation

The pYB1a-Progo-Nov1f Protein bacterial strains were cultured under induction conditions at 30°C for 16 hours. The cultures were collected, and the bacterial suspension with a OD₆₀₀ = 6 unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μL sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μL sterile water. This procedure completed the sample preparation.

2. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. Transfer the sample into the PCR instrument and heat at 100°C for 10 minutes to denature proteins. After checking for leaks with the buffer

solution, mix 32 μ L of prepared sample with 8 μ L loading buffer. After that, load 12 μ L of the mixture into the gel well and add 10 μ L of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

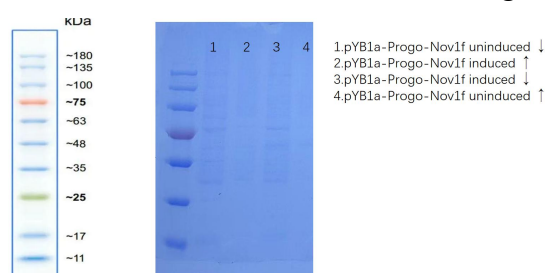
5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3.electrophoresis

Connect the power supply and run the gel at 80 V for 90 minutes

4.Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



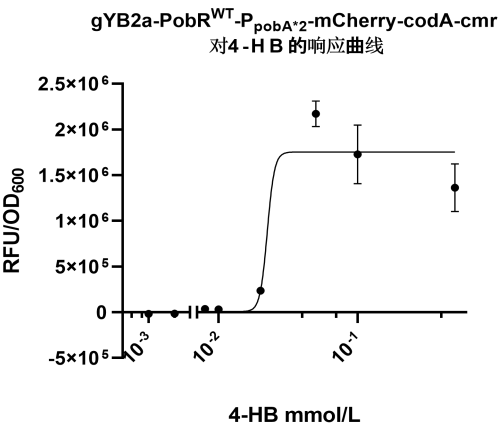
Insufficient protein samples may result from inadequate induction time or low OD₆₀₀ of harvested cells.

Module 2:

(A)Characterization of the biosensor response

1.Determination of biosensor response curve to 4- HB.

200 μL system: 1 μL bacterial culture + 0.2 μL kanamycin + 4- HB + 1 × M9. Incubate at 37 °C for 16 h.4- HB gradient: 0, 0.001, 0.002, 0.008, 0.01, 0.02, 0.05, 0.1, 0.5 mM. 6 replicates per concentration.

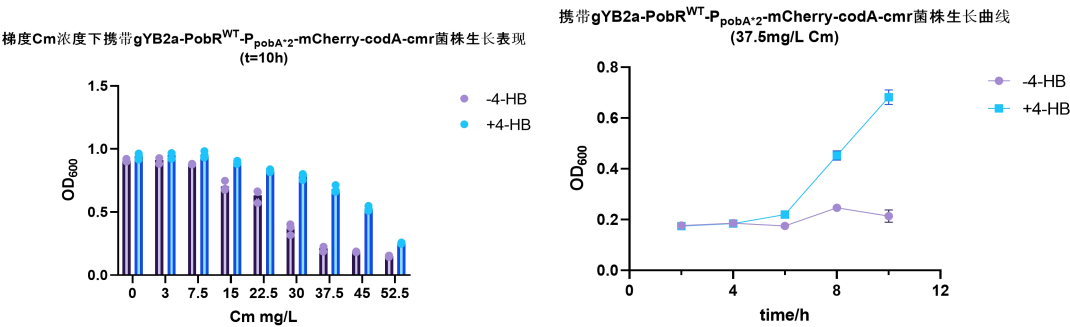


With fresh premixed system and bacterial culture, multiple replicates all exhibited a drop in the ratio at 4- HB concentrations ≥ 0.05 mM; this phenomenon has also been documented in other references.

2.Determination of saturation concentration of biosensor response to Cm

The 200 μL reaction system contained 1 μL bacterial culture, 0.2 μL kanamycin, 4- HB, LB medium and chloramphenicol. The optimal stable concentration of 4- HB was set at 0.6 mmol/L. A control group without 4- HB was also included. The samples were incubated at 37 °C for 10 h, and detection was performed every 2 h during incubation.

Chloramphenicol concentration gradient: 0, 3, 7.5, 15, 22.5, 30, 37.5, 45, 52.5 mg/L. For each chloramphenicol concentration, three replicates were prepared for both the 4- HB supplemented +4HB and 4- HB free (-4- HB) groups.



The optimal chloramphenicol concentration was 37.5 mg/L, matching expectations. The sensor showed good response to chloramphenicol.

(B)Sample preparation and HPLC detection

1.Induction:

BW, pYB1a-*eGFP* and pYB1a-*odm* strains incubated at 37 °C for 12 h were harvested. A control group without arabinose induction was set for each sample. Six tubes of ZY medium were transferred into the clean bench.

The induction system was prepared according to the following formula: 100 µL of 50×M stock, 100 µL of 5052, 10 µL of magnesium sulfate, 10 µL of 1000× stock, 5 µL of antibiotics, 50 µL of bacterial culture, and 50 µL of 20% arabinose inducer.

Induction was performed in a shaker at 30 °C for 16 h.

2.Whole-cell catalysis:

Measure OD with spectrophotometer, normalize cell concentration to $OD_{600}=6$. Formula: sampling volume (mL) = $6 / \text{measured } OD_{600}$. Aliquot bacterial culture into 1.5 mL tubes as calculated.

Balance tubes strictly, centrifuge at 4200 rpm for 10 min, discard all supernatant.

Prepare 200 µL reaction system: vanillic acid at final concentrations of 2.5 mM and 5 mM, solvent: 50 mM Tris-HCl buffer (pH 7.5). Add to cell pellets.

Set one 0 h control per group (12 samples total), store at -20 °C. Catalysis groups: shake at 30 °C, 200 r/min, protected from light for 12 h.

3.Sample preparation:

All samples after catalytic reaction were collected and centrifuged at 13000 rpm for 10 min to fully pellet bacterial cells.

The clear supernatant was carefully aspirated.

The supernatant was filtered through a 0.22 µm aqueous-phase membrane filter to remove impurities, and the filtrate was collected in a collection vial.

The filtrate was transferred into new HPLC inserts, labelled with sample information, and placed into reusable HPLC sample vials.

All prepared samples were stored at -20 °C in a refrigerator before HPLC analysis.

4.Mobile phase preparation

Prepare mobile phase: 0.1% phosphoric acid aqueous solution : methanol = 75:25, mix well. Prepare 10%

methanol and pure methanol, transfer into large bottles and label with name and date.

This is water-organic mixed mobile phase. Vacuum filter the aqueous phase into HPLC solvent reservoir to remove fine impurities.

After filtration, sonicate for 30 min for degassing to remove dissolved bubbles, avoiding unstable baseline and spurious peaks.

5.HPLC conditions and results

Mobile phase: 0.1% aqueous phosphoric acid : methanol = 75:25

Column: C18, 250 mm × 4.6 mm, 5 μm

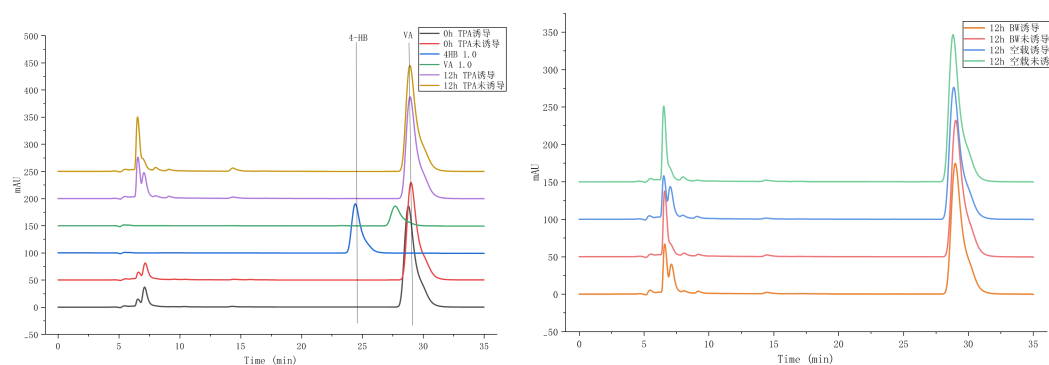
Detection wavelength: 254 nm

Injection volume: 5 μL

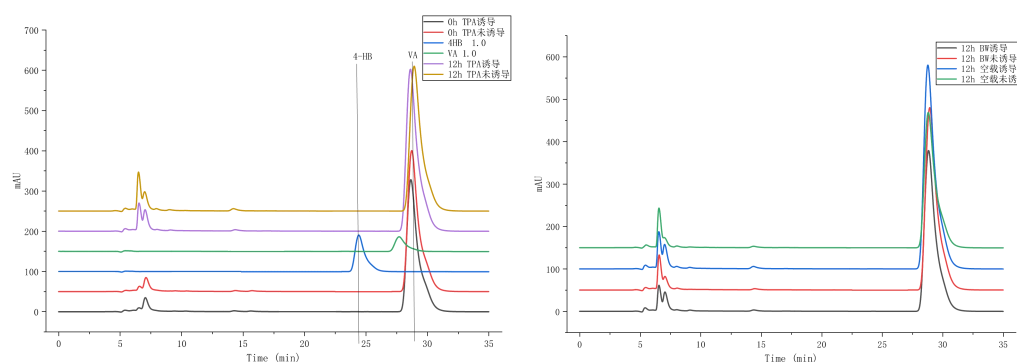
Flow rate: 0.5 mL/min

Run time per sample: 35 min

The catalytic mixture contained vanillic acid with a final concentration of 2.5 mM, using 50 mM Tris- HCl buffer (pH 7.5) as solvent.



ODM induction group: A peak appeared at approximately 24.3 min with a peak area of 5.7. Observed uncertainty: A peak was also detected at around 24.3 min in the uninduced ODM group, with a peak area of 4. The catalytic mixture contained vanillic acid with a final concentration of 5 mM, using 50 mM Tris- HCl buffer (pH 7.5) as solvent.



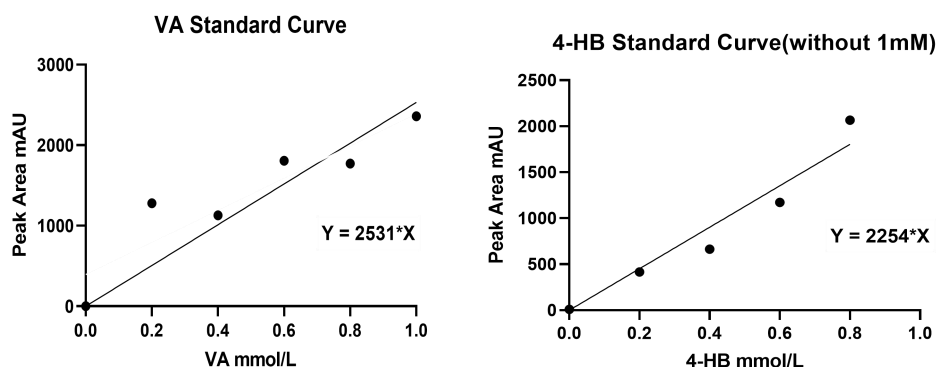
The difference was not as prominent as that in the 2.5 mM group. ODM induced: Peak at ~24.3 min, peak area = 4.1. ODM uninduced: Peak also appeared about 24.3 min, peak area = 3.2.

Add control: Add empty conversion solution (50 mM Tris- HCl, pH 7.5) to TPA induced group.

Add control: After 12 h catalysis of TPA induced group, spike 10 μL 1 mM 4- HB solution, to check whether the peak at 24.3 min is 4- HB. Higher peak area indicates the peak is 4- HB.

Prepare whole-cell catalytic system with 2.5 mM vanillic acid substrate.

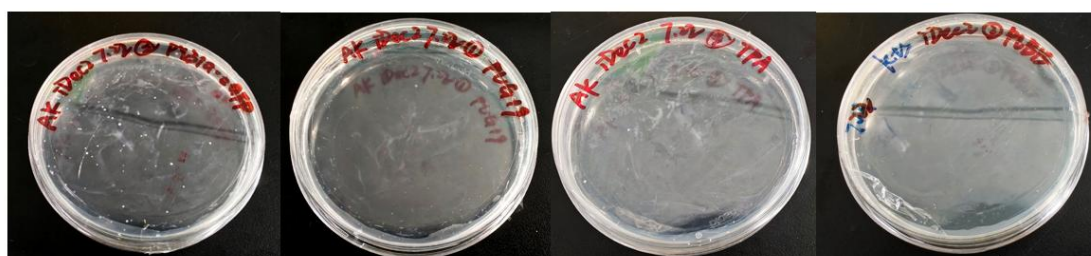
6. Standard curve



Issue: Poor standard curve fitting. Suspected problematic solution preparation, but system and dilution process were confirmed correct. Next step: re-prepare stock solution and perform re-dilution.

(C) Competent cell preparation and validation results

1. Validation results

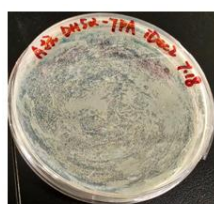


PobR transformed with empty pYB1a plasmid

PobR transformed with pUC19 validation plasmid. Colony count: 150.

Transform PobR with freshly extracted mixed TPA plasmids.

Transform PobR with pLB1s-aroY plasmid.



Transform DH5α with mixed ODM plasmids.

Based on validation results: the problem originated from PobR competent cell preparation. Next preparation: culture bacteria at 30 °C for 1 h, then shift to 16 °C for 3–4 h.

2. Transformation

The mixed plasmid concentration was 30 ng/μL.

After the competent cells were incubated on ice for 5 min, 1 μL of mixed plasmid was added. The mixture was kept on ice for 20 min, followed by heat shock at 42 °C for 45 s, and immediately placed back on ice for 2–3 min. Subsequently, 600 μL LB medium was added to the EP tube, and the suspension was cultured at 37 °C in

a shaker for 45 min.

The tube was centrifuged at 4200 r/min for 2 min.

100 μ L of the suspension was taken in the clean bench, mixed gently, and spread onto agar plates.

3.pUC19 validation plasmid transformation

After the competent cells were incubated on ice for 5 min, 1 μ L of pUC19 validation plasmid 10pg/ μ L was added.

The mixture was kept on ice for 25 min, subjected to heat shock at 42 $^{\circ}$ C for 45 s, and immediately placed back on ice for 2–3 min.

600 μ L LB medium was added to the EP tube, and the sample was shaken at 37 $^{\circ}$ C for 45 min.

Centrifugation was performed at 4200 r/min for 2 min. 100 μ L of bacterial suspension was aspirated in the clean bench, mixed gently and spread onto agar plates.

Module 3:

(A)Run HPLC to verify pyruvate production.

1.Shake culture of strains No.1, 2 and 12, and send for testing

Strains No.1, No.2 and No.12 were inoculated and cultivated in LB medium (37 $^{\circ}$ C, 12 hours), and then submitted for sequencing/testing.

2.Culture preservation and plasmid extraction

The strains were preserved (glycerol stock), and plasmids were extracted and their concentrations were measured.

3.Induction: Induction of BW-pLB1s-*VanAB-aroY*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO4	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30 $^{\circ}$ C for 16 hours.

4.Whole-cell catalysis

Measurement of the OD600 value.

BW-pYB1a-dmpB-dmpD-mhpD-mhpE using VA to catalyze the induced enzymes for Pyr production. BW-pLB1s-VanAB-aroY, BW-pYB1a-dmpB-dmpD-mhpD-mhpE using CA to catalyze the induced enzymes for Pyr production. Determine the OD 600 of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD 600=6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in H2SO4, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30 ° C for 1 hours.

The whole-cell catalytic system was configured as follows:

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	/	5mM H2SO4 solution
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE))	VA 5mM (final)	/	5mM H2SO4 solution
Induced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE))	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced BW	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced BW	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a-eGFP	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced BW-pYB1a-eGFP	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	/	5mM H2SO4 solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	/	5mM H ₂ SO ₄ solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution
Induced BW	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW-pYB1a-eGFP	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution
Induced BW-pYB1a-eGFP	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution

All components were added to the cell pellet tubes and mixed by pipetting, then shaken at 30°C, 200r/min in the dark.

5.Prepare standards and mobile phase

The HPLC standard solutions and the mobile phase (5mM H₂SO₄ aqueous solution, etc.) were prepared.

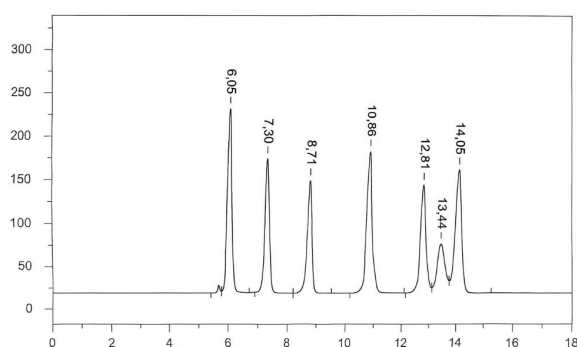
6.Rinse and activate the new column (Aminex HPX-87H)

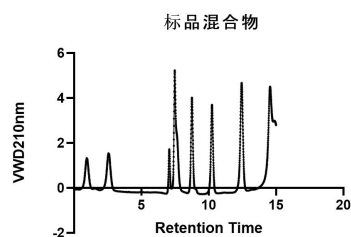
The new column (Aminex HPX-87H) was rinsed and activated.

Mobile phase: 5mM H₂SO₄ aqueous solution; 5% acetonitrile/5mM H₂SO₄ and 30% acetonitrile/5mM H₂SO₄ were also prepared. Flow rate: 0.6 mL/min; column temperature: 50°C; column: Aminex HPX-87H; injection volume: 10μL.

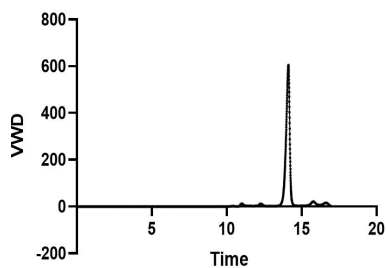
The pyruvate standards (10/20/30 mM, peak areas close to 1:2:3) and the full-pathway and downstream-pathway samples were detected by HPLC.

For the full-pathway samples, an obvious consumption of vanillic acid was preliminarily judged; the downstream-pathway samples were also verified.

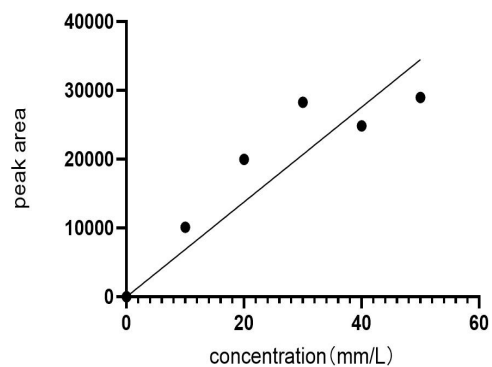
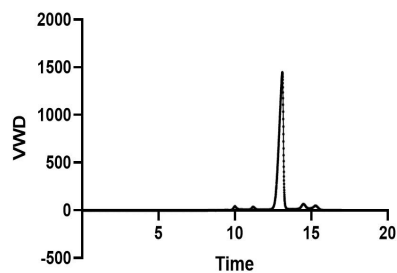




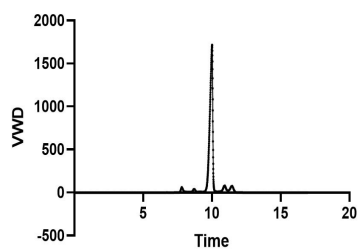
10mMpyr



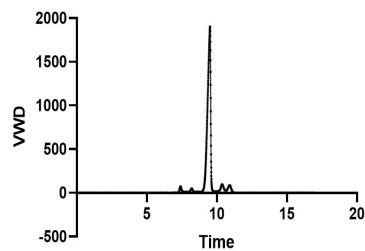
30mMpyr



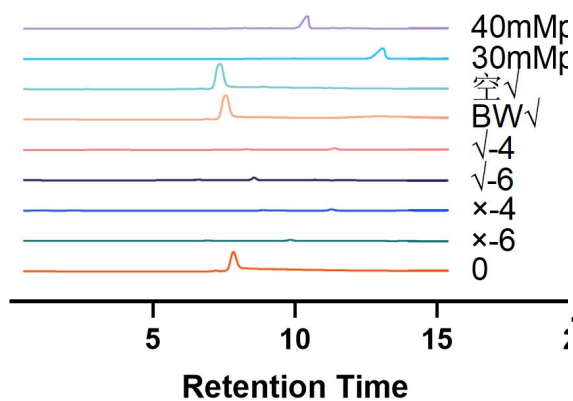
40mMpyr



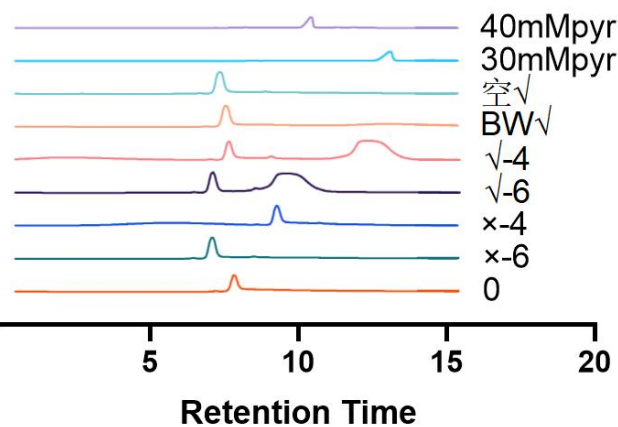
50mMpyr



A-Y



A-Y + NADH



7.HPLC run.

The standards and the samples (full pathway and downstream pathway) were detected by HPLC with RID.

8.HPLC result analysis (peak shifts due to unstable temperature control)

Because the column oven was not equipped with a heat-insulation box, the temperature was unstable and the peak times shifted. For the full-pathway samples, an obvious consumption of vanillic acid was preliminarily judged, but the pyruvate product peak could not be confirmed due to the temperature-control problem.

(B)Experimental design and 96-well plate variable setting

96-well addition system for full-pathway and downstream-pathway verification.

A-E: the full-pathway strain (pyruvate sensor + pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE); B-E: the downstream-pathway strain (pyruvate sensor + pYB1a-dmpBD-mhpDE); empty vector: pYB1a-eGFP. Culture volume 2μL per well.

Full-pathway and downstream-pathway groups (per well):

No.	Group	Medium	Additives
1	Induced A-E, blank	198μL plain 1×M9 (AS)	—
2	Induced A-E + vanillic acid	198μL 1×M9 with vanillic acid (AS)	—
3	Induced A-E + vanillic acid + NADH	190μL 1×M9 with vanillic acid (AS)	8μL NADH stock
4	Induced A-E + sodium pyruvate (positive)	158μL plain 1×M9 (AS)	40μL NaPyr stock
5	Uninduced A-E, blank	198μL plain 1×M9 (AS)	—
6	Uninduced A-E + vanillic acid + NADH	190μL 1×M9 with vanillic acid (AS)	8μL NADH stock
7	Induced eGFP, blank	198μL plain 1×M9 (A)	—
8	Induced eGFP + vanillic acid	198μL 1×M9 with vanillic acid (A)	—
9	Induced eGFP + vanillic acid + NADH	190μL 1×M9 with vanillic acid (A)	8μL NADH stock
10	Induced eGFP + sodium pyruvate (positive)	158μL plain 1×M9 (A)	40μL NaPyr stock

No.	Group	Medium	Additives
11	Induced B-E, blank	198μL plain 1×M9 (A)	—
12	Induced B-E + catechol	188μL plain 1×M9 (A)	10μL catechol
13	Induced B-E + catechol + Fe2+/Mn2+	185μL plain 1×M9 (A)	10μL catechol, 2μL Mn2+, 1μL Fe2+ (added last)
14	Induced B-E + sodium pyruvate (positive)	158μL plain 1×M9 (A)	40μL NaPyr stock
15	Uninduced B-E, blank	198μL plain 1×M9 (A)	—
16	Uninduced B-E + catechol	188μL plain 1×M9 (A)	10μL catechol
17	Induced eGFP + catechol	188μL plain 1×M9 (A)	10μL catechol
18	Induced eGFP + catechol + Fe2+/Mn2+	185μL plain 1×M9 (A)	10μL catechol, 2μL Mn2+, 1μL Fe2+ (added last)

4. Microplate reader program setting.

The program was set as follows: temperature control at 30°C (fluctuation range 29.5-30.5°C); orbital shaking, amplitude 2.5mm, 216rpm, 4s per cycle; total running duration 23h59min59s, one round of detection every 30min; absorbance at 600nm, bandwidth 3.5, 10 flashes, 50ms settling

time before reading.

